Panchip Microelectronics Co., Ltd.

PAN3029/3060 Series Evaluation Board User Manual

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Shanghai Panqi Microelectronics Co., Ltd.

Address: Room 302, Building D, No. 666, Shengxia Road, Zhangjiang Hi-Tech Park, Shanghai Contact number: 021-50802371
Website: http://www.panchip.com

Documentation

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Revision History

Version	Revision date	Description
V1.0	2023.6	Initial version
V1.1	2023.8	Added the function of modifying LDR button
V1.2	2023.11	Updated the function description of the baseboard
V1.3	2023.12	Added the function of mode 1/2 button, added the function of single carrier test, and the power consumption test description
V1.4	2024.04	Update chip description

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1 Introduction

1.1 Overview

The PAN3029/3060 series evaluation board system includes an RF module based on the ChripIoT™ chip with independent intellectual property rights of Panqi, an HC32F460 main control integrated board, an antenna, an LCD display, and a USB Mini-B communication power cable. The evaluation system can configure the bandwidth (BW) and spreading factor (SF) of the RF module. Users can use this evaluation system to perform point-to-point communication distance, receiving sensitivity, single carrier, RF power consumption and other tests.

1.2 Configurable parameters

- Bandwidth (BW) supports 500KHz, 250KHz, 125KHz, 62.5KHz
- Spreading factor (SF) supports 5, 6, 7, 8, 9, 10, 11, 12
- Coding rate (CR) supports CR4/5, CR4/6, CR4/7, CR4/8
- Frequency (FQ)
- Transmit power (PW) 1-23 levels
- Test mode: Mode 1, Mode 2
- Transmit payload length (PL) 10-240 bytes, 10 bytes/step

2 Functional Description

The functional description of the baseboard based on HC32 is shown in Figure 2-1.

Figure 2-1 HC32 base board

2.1 Jumper Function

Jumper 1: RF module power supply jumper, where the RF operating current can be measured.

Jumper 2: HC32F460 module power supply jumper, where the MCU operating current can be measured.

2.2 Button Functions

There are 6 buttons on the RF evaluation board, and the specific button definitions are as follows:

- **Button 1:**
  Function selection button. If button 1 is not pressed, press other buttons to select the first function of other buttons; while button 1 is pressed, press other buttons to select the second function of other buttons.
- **Button 2:**
  The first function is to clear the statistics of the sent and received packets, and the statistics are displayed in the lower right corner (TX or RX);
  The second function is to switch between test mode 1 and test mode 2. After entering, '1' and '2' are displayed in the upper right corner of the screen. The test mode is not entered by default.
- **Button 3:**
  The first function is to start sending (only valid in the transmission mode). Only a single press is required. The transmission status of each press needs to be used in conjunction with button 4;
  The second function is to switch the frequency value, which is displayed on the screen FQ in units of 0.1MHz.
- **Button 4:**
  The first function is to select the continuous transmission mode. The configuration is valid only in TX mode (the DIP switch is turned up to TX). The screen upper right corner displays (MODE). The available modes are A/B/C/D. Mode A means sending a single packet, mode B means sending 100 packets continuously, mode C means sending 9999 packets continuously, and mode D means entering the single carrier test mode. In the single carrier test mode, the transmission frequency and transmission power can be modified by combining the buttons;
  The second function is to switch CodeRate. The values that can be set are 4/5, 4/6, 4/7, and 4/8.
- **Button 5:**
  The first function is to set SF. The values that can be set are 5/6/7/8/9/10/11/12;
  The second function is to set the payload length (10~240, step 10).
- **Button 6:**
  The first function is to set BW. The values that can be set are 500K/250K/125K/62.5K;
  The second function is to set the transmit power. The screen PW displays the power level. The value is a decimal value. For the corresponding transmit power, refer to the power table appendix of the SDK user guide.

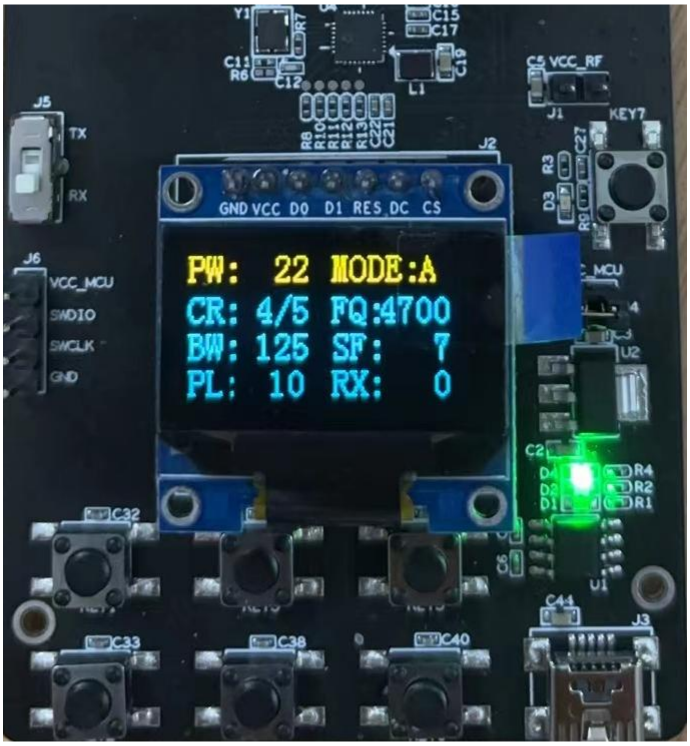

Figure 2-2 Evaluation board OLED display

2.3 Interface Description

Rod antenna interface: connect rod antenna.
Serial port: used to power the baseboard and debug the program, the default baud rate of the program is 115200.
Power supply: powered by USB Mini-B power cable (5V).

2.4 Reset button

Reset button: Press the reset button to reset HC32F460.

2.5 DIP switch

Dip switch: up to TX means the module is set to transmit mode, down to RX means the module is set to receive mode. Switching the module's transmit and receive modes by DIP code will take effect after resetting HC32F460.

2.6 Point-to-point communication distance test

Use two evaluation boards, turn the DIP switch of one evaluation board to TX state and the other to RX state, set the RF parameters of the two evaluation boards to the same (frequency, SF, BW), adjust the transmit power level of the transmitter, select the corresponding transmit mode by pressing button 4, trigger the send by pressing button 3, and check the RX reception success count on the receiving end.

2.7 Mode 1/2 (receiving sensitivity test)

The second function of button 2 is to switch between test mode 1 and test mode 2. After entering, ‘1’ and ‘2’ are displayed in the upper right corner of the screen. Test mode 1 can perform sensitivity test. Test mode 1 uses the TestModeBuffer (default value is ten bytes of 0x00) and TEST_MODE_BUFFER_LEN (default value 10) data defined in the OLED DEMO software for testing. If the length of the data received by the module is TEST_MODE_BUFFER_LEN, and the data content is correctly compared with TestModeBuffer, the RX count is increased by 1. Test mode 2 does not perform software data comparison. If the module receives RXDONE, the RX count is increased by 1.

The default configuration does not enter the test mode, and ‘1’ or ‘2’ is not displayed in the upper right corner of the screen, indicating normal data communication. The RX count also only indicates the packet receiving count.

2.8 Continuous wave test

For single carrier test, first turn the EVB DIP switch upward to TX and then power on again. Use the buttons to configure the relevant RF parameters, and press button 4 to switch the mode to D mode - single carrier test mode. Then press button 3 to trigger the single carrier transmission function.

2.9 Power consumption test

The receiving and transmitting currents of the RF module can be measured through jumper cap 1. If you want to measure the power consumption performance related to sleep, you can burn the AT DEMO project. Please refer to the AT command reference document and send AT commands to make the RF module enter the relevant state for testing.